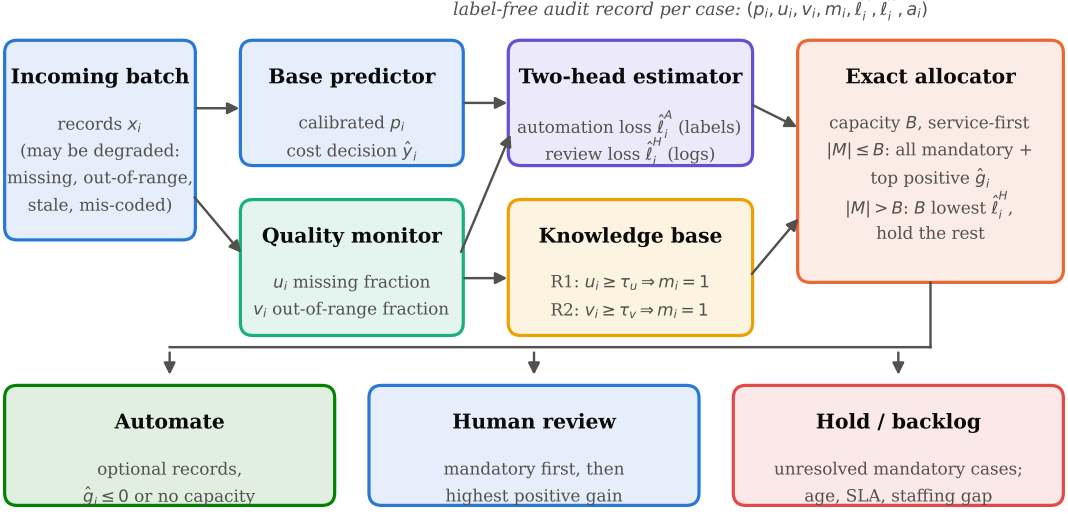
label-free audit record per case: $(p_i, u_i, v_i, m_i, \hat{\ell}_i^A, \hat{\ell}_i^H, a_i)$
Incoming batch
records x_i
(may be degraded: missing, out-of-range, stale, mis-coded)
Base predictor
calibrated p_i
cost decision $\hat{y}_i$
Quality monitor
u_i missing fraction
v_i out-of-range fraction
Two-head estimator
automation loss $\hat{\ell}_i^A$ (labels)
review loss $\hat{\ell}_i^H$ (logs)
Knowledge base
R1: $u_i \geq \tau_u \Rightarrow m_i = 1$
R2: $v_i \geq \tau_v \Rightarrow m_i = 1$
Exact allocator
capacity B, service-first
$|M| \leq B$: all mandatory + top positive $\hat{g}_i$
$|M| > B$: B lowest $\hat{\ell}_i^H$, hold the rest
Automate
optional records, $\hat{g}_i \leq 0$ or no capacity
Human review
mandatory first, then highest positive gain
Hold / backlog
unresolved mandatory cases; age, SLA, staffing gap